Record-Level Evaluation of a Morlet CNN–Conformer for Five-Class ECG Beat Classification

Alireza Abbaszadeh[a,*], Vahid Torkzadeh[a]

[a]*Department of Computer Engineering, Mashhad Branch, Islamic Azad University, Ostad Yousefi St., Ghasem Abad, Mashhad, 91871, Iran*

Abstract

Introduction and Objectives: Class imbalance and reuse of a patient's beats during model development and testing can inflate ECG classification performance. We evaluated a Morlet CNN–Conformer on a record-level holdout and examined the errors underlying its summary scores. **Methods:** Forty-five MIT–BIH records were divided into a 38-record development set and a seven-record test set. Reference annotations centred 187-sample segments from the first ECG channel. Magnitudes of a 32-scale Morlet continuous wavelet transform were grouped into three-beat sequences. A shared convolutional beat encoder followed by a Conformer was compared with CNN–attention, CNN–LSTM, and CNN-only models. **Results:** The test set contained 15,573 sequences. The CNN–Conformer achieved an accuracy of 0.60 (95% CI 0.59–0.60), a macro-F1 of 0.26 (0.26–0.27), and a weighted-F1 of 0.68 (0.67–0.68), the highest values among the four models. F1 was 0.74 for Normal and 0.57 for ventricular ectopic beats, but 0.002 for supraventricular ectopic beats and 0.007 for Fusion beats. Most supraventricular ectopic beats were classified as Normal, whereas most Fusion beats were classified as ventricular ectopic. **Conclusions:** The Conformer-based sequence encoder improved aggregate performance within this experiment, but the gain was carried mainly by the two common classes. Longer rhythm context, the second lead, and development-set selection of minority-class strategies should be tested before external evaluation.

Keywords: electrocardiography, heartbeat classification, record-level validation, Conformer, continuous wavelet transform, class imbalance

1. Introduction

An ECG classifier can appear convincing while missing the rhythms for which automated assistance may matter most. Normal beats usually account for most observations in ambulatory recordings, whereas supraventricular ectopic beats (SVEB) and fusion beats are comparatively uncommon and unevenly distributed among patients. Accuracy and weighted averages largely follow the common classes under these conditions. Macro-averaged measures and precision–recall curves expose a different part of

*Corresponding author

Email addresses: `alireza.abbaszadeh8558@iau.ir` (Alireza Abbaszadeh), `torkzadeh@mshdiau.ac.ir` (Vahid Torkzadeh)

the result, but they are still meaningful only when class support and the construction of the test cohort are reported[1, 2]. Recent ECG studies have reached the same problem through different clinical tasks, including signal classification, atrial-fibrillation detection, and rhythm discrimination in clinical recordings[3, 4, 5].

Patient separation is particularly important in beat classification. Beats from one recording share morphology, acquisition conditions, and rhythm history; random allocation of individual beats can therefore place closely related examples on both sides of the training–test split. Holding out entire records gives a more demanding estimate of generalisation, although the estimate may vary sharply when only a few records contain the rare classes[6]. This dependence on population and acquisition setting is also evident in studies of automated 12-lead interpretation, mobile lead-I risk assessment, and single-lead QT monitoring[7, 8, 9]. Results from those tasks are not numerical benchmarks for MIT–BIH beat classification, but they reinforce the need to define exactly what was held out and what each reported metric represents.

The model considered here combines a time–frequency description of individual beats with short-range rhythm context. Wavelet coefficients retain transient ECG morphology at several scales[10, 11, 12]; related work has used multiresolution representations for ECG denoising and explicit feature selection for rhythm detection[13, 4]. Context across beats is a separate source of information, as illustrated by analyses based on long-term RR dynamics[14]. We used two-dimensional convolutions to encode Morlet scalograms and a Conformer block to model a sequence of three beats[15]. The Conformer was chosen because its attention and convolutional modules can represent interactions across the short sequence, not because either module has already been shown to solve the minority-class problem.

2. Objectives

The primary objective was to evaluate a Morlet CNN–Conformer on ECG records kept outside model development and to compare its aggregate performance with CNN–attention, CNN–LSTM, and CNN-only models sharing the same convolutional beat encoder. A secondary objective was to characterize class-specific errors, particularly for the minority SVEB and Fusion classes, so that the limits of the current experiment are explicit before broader claims of generalisation or clinical usefulness are made.

3. Materials and Methods

3.1. Study design and data source

This retrospective computational study used the MIT–BIH Arrhythmia Database, a collection of 48 half-hour, two-channel ambulatory ECG recordings sampled at 360 Hz[16, 17]. The analysis included 45 records; records 102, 104, and 107 were excluded. The remaining records were divided before model fitting. Records 106, 207, 208, 220, 228, 231, and 233 formed the test set, and the other 38 records formed the development set. The split was generated at record level with a fixed random seed of 42. No beat sequence from a test record was used for hyperparameter search, cross-validation, normalisation, or model fitting.

The development records were used in two stages. Hyperparameters for each architecture were selected on the first training–validation partition of a shuffled five-fold record split. The selected CNN–Conformer configuration was then assessed across all five development folds. Finally, each selected architecture was fitted to all 38 development records and evaluated once on the same seven test records.

3.2. Beat representation

Reference annotations supplied with the database defined the centre of each beat; no separate QRS detector was used. From the first ECG channel, we extracted 93 samples before and after the annotated location together with the centre sample, giving a 187-sample segment (approximately 519 ms). Segments extending beyond a recording boundary and annotations not represented in the study label map were omitted.

For every retained segment, a continuous wavelet transform was calculated with the Morlet wavelet at integer scales 1–32. The absolute values of the coefficients formed a 32×187 scalogram. Original beat symbols were grouped into five AAMI-oriented classes: Normal (N), supraventricular ectopic beat (S), ventricular ectopic beat (V), Fusion (F), and Unknown or paced (Q)[6]. Within each record, overlapping sequences of three consecutive scalograms were formed; the class of the third beat was the prediction target. Means and standard deviations were estimated from the relevant training records at each of the 187 sample positions and then applied to the corresponding validation or test scalograms. This procedure was repeated within each cross-validation fold and used all 38 development records for the final models.

3.3. Model architectures

All four models began with the same beat encoder. Each scalogram passed through three 3×3 convolutional layers with 32, 64, and d filters. Batch normalisation and a rectified linear activation followed each convolution; the first two layers also used 2×2 max pooling. Global average pooling reduced each encoded scalogram to a d-dimensional vector, leaving a sequence of three vectors for classification.

In the proposed model, this sequence entered a Conformer block containing two feed-forward modules, multi-head self-attention, and a depthwise convolutional module, with residual connections and layer normalisation[15, 18, 19, 20]. The selected configuration used $d = 128$, eight attention heads, a feed-forward expansion factor of four, a convolution kernel of 11, and dropout of 0.2. The three output vectors were averaged before the five-class softmax layer.

The comparison models differed only after the shared convolutional beat encoder. The CNN–attention model used two attention/feed-forward blocks without the Conformer convolutional module ($d = 64$, eight heads, expansion factor four, no dropout). The CNN–LSTM used a 128-unit LSTM after a 128-dimensional beat encoder. In the CNN-only model, three 128-dimensional beat embeddings were averaged directly before classification. These comparisons were not treated as component-wise ablations because the selected embedding dimensions and number of sequence blocks were not identical across architectures.

3.4. Training and model selection

Hyperband searches were conducted separately for the four architectures, with a maximum of 20 epochs per trial[21]. The search considered an embedding dimension of 64 or 128; architecture-specific ranges included four or eight attention heads, a feed-forward expansion factor of two or four, one or two sequence blocks, and dropout up to 0.3. Trials were ranked by validation accuracy. Validation loss governed early stopping and learning-rate reduction during the searches.

All final models were trained for 20 epochs with a batch size of 128. Optimisation used AdamW with a learning rate of 1×10^{-5} and sparse categorical cross-entropy. Class weights were calculated from the labels in the development set to give equal total weight to each observed class. The same record split, input representation, optimisation rule, and test cohort were used for every architecture.

3.5. Performance measures and statistical analysis

We calculated accuracy, macro-F1, weighted-F1, and class-specific precision, recall, and F1. One-vs-rest receiver operating characteristic area under the curve (AUC) was used to describe ranking discrimination, and precision–recall curves were retained because of the marked class imbalance[22, 2]. The two Q observations in the test set were considered insufficient for a class-specific AUC estimate. Development-fold accuracy is summarised by its mean and standard deviation.

Confidence intervals for test-set accuracy and F1 were calculated from 1,000 nonparametric bootstrap samples of the 15,573 test sequences, using the 2.5th and 97.5th percentiles. The same procedure was used for class-specific AUC when both outcomes were present in a resample. These beat-level intervals quantify sampling variation among the observed sequences; they do not account for correlation within records or uncertainty in selecting a new patient cohort.

After the principal evaluation, thresholds for S and F were varied in one-vs-rest analyses of the test probabilities. We also calculated 15-bin expected calibration error and Brier scores and fitted a single temperature parameter to the same probabilities. Because neither thresholds nor temperature were selected on independent data, these analyses are reported as descriptive diagnostics rather than validated operating points.

Table 1: Distribution of prediction targets in the held-out test set.

AAMI class	Number	Percentage
Normal (N)	12,061	77.5
SVEB (S)	214	1.4
VEB (V)	2,913	18.7
Fusion (F)	383	2.5
Unknown/paced (Q)	2	< 0.1
Total	15,573	100.0

Table 2: Aggregate performance on the common record-level test set. Values in brackets are 95% percentile intervals from 1,000 beat-level bootstrap samples.

Model	Accuracy (95% CI)	Macro-F1 (95% CI)	Weighted-F1 (95% CI)
CNN–Conformer	**0.597 [0.590, 0.605]**	**0.263 [0.260, 0.267]**	**0.677 [0.671, 0.684]**
CNN–attention	0.269 [0.262, 0.277]	0.149 [0.145, 0.153]	0.379 [0.370, 0.387]
CNN–LSTM	0.179 [0.173, 0.185]	0.154 [0.149, 0.159]	0.286 [0.278, 0.295]
CNN-only	0.136 [0.131, 0.142]	0.069 [0.066, 0.073]	0.226 [0.218, 0.233]

4. Results

4.1. Test cohort

The seven test records yielded 15,573 three-beat sequences. Normal beats accounted for 12,061 targets (77.5%) and ventricular ectopic beats (VEB) for 2,913 (18.7%). Only 214 targets were SVEB (1.4%) and 383 were Fusion beats (2.5%); two belonged to Q (Table 1). Accuracy for the CNN–Conformer across the five development folds ranged from 0.21 to 0.64, with a mean of 0.44 and standard deviation of 0.16.

4.2. Comparison of the four models

The CNN–Conformer produced the highest aggregate scores on the common test set (Table 2; Figure 1). Its accuracy was 0.60 (95% CI 0.59–0.60), macro-F1 was 0.26 (0.26–0.27), and weighted-F1 was 0.68 (0.67–0.68). The corresponding values for the CNN–attention model were 0.27, 0.15, and 0.38. The CNN–LSTM reached an accuracy of 0.18 and the CNN-only model 0.14. Although the CNN–Conformer ranked first on all three aggregate measures, it did not have the highest AUC for every class; the CNN–LSTM VEB AUC, for example, was 0.86 compared with 0.84 for the CNN–Conformer.

4.3. Class-specific performance

The aggregate result was not shared evenly across the five classes. Normal beats had a precision of 0.95, a recall of 0.60, and an F1 of 0.74. For VEB, precision was 0.48, recall 0.70, and F1 0.57. In contrast, the model correctly identified one of 214 SVEB targets (F1 0.002 before rounding) and four of 383 Fusion targets (F1 0.007). AUC was 0.90 for Normal, 0.84 for VEB, 0.67 for SVEB, and 0.31 for Fusion (Table 3; Figure 2). No inferential class-level result was assigned to Q because the test set contained only two observations.

The confusion matrix showed two dominant minority-class errors. Of 214 SVEB targets, 151 were classified as Normal and 51 as VEB. Of 383 Fusion targets, 343 were classified as VEB (Figure 3).

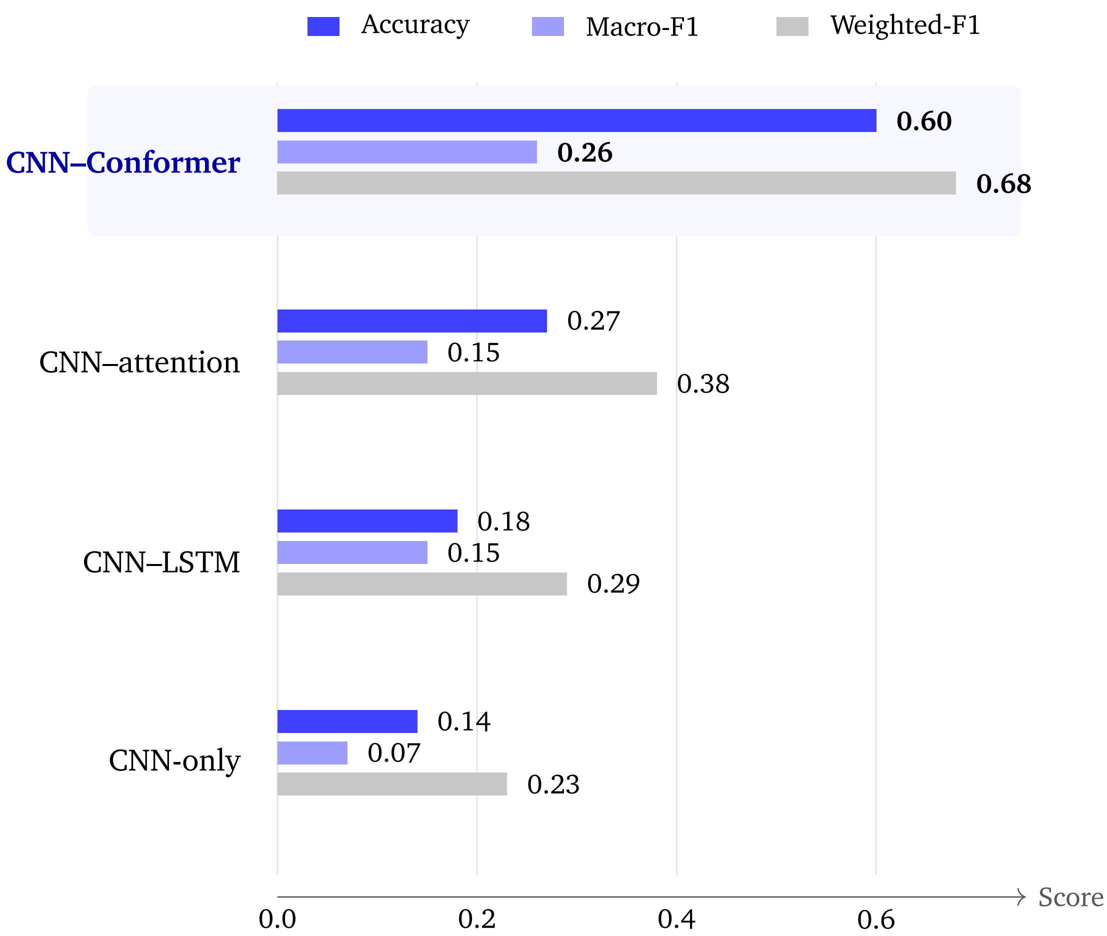

Figure 1: Aggregate performance of the four architectures on the same seven test records. Error intervals are given in Table 2.

4.4. Exploratory analyses

Changing the decision threshold did not produce a balanced operating point for either rare class. The largest observed SVEB F1 was 0.06 at a threshold of 0.09 (precision 0.03, recall 0.90). For Fusion, the largest F1 was 0.05 at a threshold approaching zero (precision 0.02, recall 1.00). These values were calculated on the test set and are descriptive.

Temperature scaling produced a similarly small change. The fitted temperature was 1.050, and mean negative log-likelihood changed from 1.047989 to 1.047731. Class-specific calibration results are provided in the Supplementary Material; because the temperature was fitted on the test probabilities, they are not estimates of calibration on new data.

Table 3: Class-specific performance of the CNN–Conformer on the held-out test set.

AAMI class	Support	Precision	Recall	F1	AUC (95% CI)
Normal (N)	12,061	0.948	0.603	0.737	0.901 [0.89, 0.91]
SVEB (S)	214	0.001	0.005	0.002	0.668 [0.65, 0.68]
VEB (V)	2,913	0.483	0.696	0.570	0.843 [0.83, 0.85]
Fusion (F)	383	0.006	0.010	0.007	0.314 [0.29, 0.34]
Unknown/paced (Q)	2	0.000	0.000	0.000	—

AUC was not interpreted for Q because only two Q targets were present. Bootstrap intervals resampled beats and do not account for clustering within records.

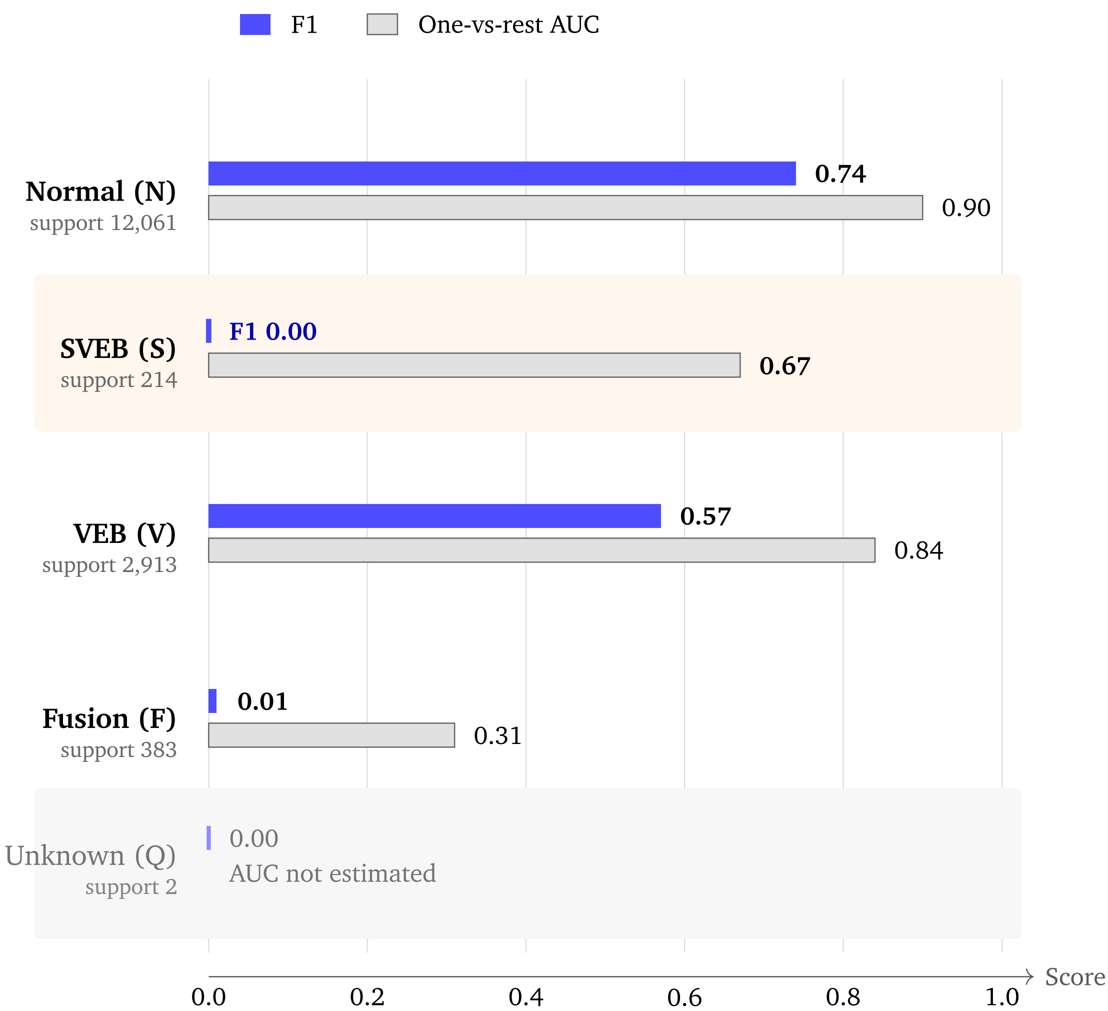

Figure 2: F1-score and one-vs-rest AUC for the CNN–Conformer, shown with the number of test observations in each class. AUC is not displayed for Q because only two Q observations were available.

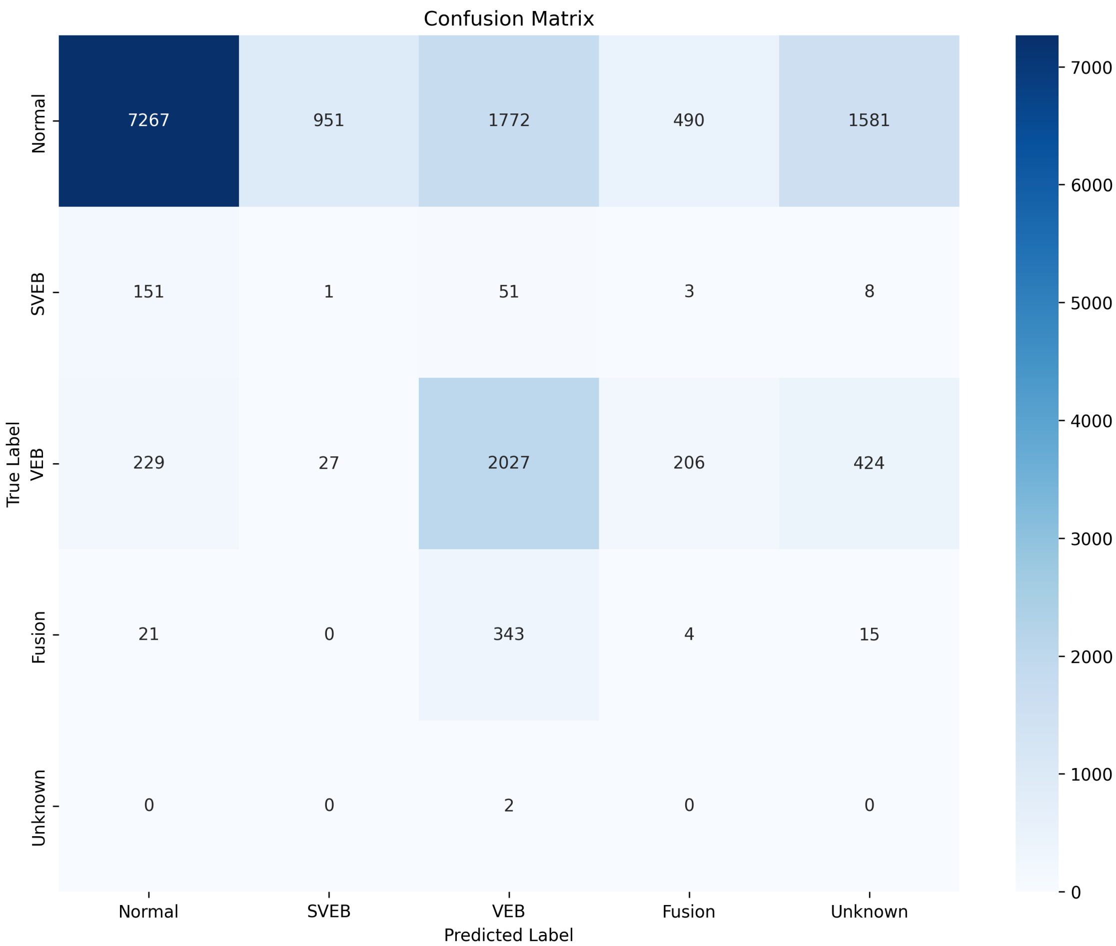

Figure 3: Confusion matrix for the CNN–Conformer on the record-level test set. Rows show reference classes and columns show predicted classes.

5. Discussion

On the seven records withheld from development, the CNN–Conformer outperformed the three comparison models on accuracy, macro-F1, and weighted-F1. The ranking is clear within this experiment, but its scope is narrow. Normal and VEB were the only classes with useful F1-scores; SVEB and Fusion were almost never recognised. Describing the model simply as the best of the four would therefore miss the more important finding: adding a Conformer sequence encoder improved the aggregate ranking without producing a dependable five-class classifier.

The gap between weighted-F1 (0.68) and macro-F1 (0.26) follows directly from the composition of the test set. Normal and VEB made up 96.2% of its targets, so success on those classes dominates the weighted result. Macro-F1 gives the rare classes equal influence and falls accordingly. This is not merely a preference among summary statistics. In an arrhythmia classifier, a high weighted score and near-zero F1 for SVEB and Fusion describe clinically different capabilities, even though both statements arise from the same predictions.

The comparison among architectures needs equally careful interpretation. Every model used the same convolutional encoder for individual scalograms. The CNN–attention comparator differed from the proposed model in its sequence encoder, but it also had a smaller embedding and two blocks rather than one. The CNN–LSTM and CNN-only models had their own selected configurations. They therefore provide model-level comparisons on common data, not a controlled removal of one Conformer component. The results support the selected CNN–Conformer as the strongest of the four fitted systems; they do not identify self-attention, the Conformer convolution, or their interaction as the cause of the difference.

The errors suggest where a more informative experiment should begin. Seventy-one per cent of SVEB targets became Normal, whereas 90% of Fusion targets became VEB. The first pattern is compatible with a representation that gives too little weight to rhythm timing, which may not be adequately conveyed by only three beats. The second is compatible with ventricular morphology dominating a fusion beat's representation. Both explanations are physiologically plausible, but neither can be inferred from a confusion matrix alone. Varying sequence length and lead count while holding the encoder and training set fixed would test them more directly.

The threshold analysis makes a simple operating-point explanation unlikely. Recovering most SVEB or Fusion targets required accepting precision of 0.03 or 0.02. In other words, the rare-class scores overlapped extensively with scores for other beats; moving a threshold mainly exchanged false negatives for many false positives. Global temperature scaling also had negligible influence on negative log-likelihood. Work on certainty in AI-enabled ECG and calibrated uncertainty for single-lead devices has shown why discrimination and confidence should be examined separately[23, 9]. Here the calibration analysis was fitted and read on the same test set, so it can only indicate that one global correction did not repair the observed class-specific failures.

The broader ECG literature offers relevant design choices rather than directly comparable performance targets. Deep networks have been studied for signal classification and atrial-fibrillation detection[3, 5]; other work has emphasised selected ECG features, long-term RR dynamics, or mobile recordings[4, 14, 8]. Those studies differ in endpoint, signal duration, and cohort construction. The present results add a more limited observation: with a Morlet representation, three-beat context, and record-level separation, the largest remaining errors were concentrated in specific physiologically plausible class transitions.

Several features of the study limit that observation. The data come from one historical database, and the final estimate rests on seven test records. Beat-level bootstrap intervals are consequently narrower than intervals that would incorporate variation among patients. The analysis excluded records 102, 104, and 107 but retained record 217, which includes paced annotations mapped to Q; the cohort should not be interpreted as completely free of paced recordings. Only the first channel and three adjacent beats were analysed, the test set contained two Q targets, and class weighting was the only imbalance strategy used in final training. The threshold and calibration analyses reused the test probabilities and cannot be carried forward as validated settings.

A useful next experiment is therefore smaller and more controlled than another broad architecture search. Sequence length and use of the second channel should first be varied on the existing development records, with SVEB-to-Normal and Fusion-to-VEB errors specified in advance as outcomes. Focal or class-balanced objectives can then be compared with the current weighted cross-entropy, while thresholds remain selected inside the development set. Only after those choices are fixed would external evaluation be informative. Contemporary 12-lead interpretation and ECG digitisation studies also show that acquisition and signal quality can materially affect downstream performance[7, 24]; an external cohort should therefore differ not only in patients but, where possible, in acquisition conditions.

6. Conclusions

The Morlet CNN–Conformer gave the best aggregate results among four models tested on the same held-out records, but it recognised almost none of the SVEB or Fusion targets. Its apparent advantage therefore reflects improved performance on Normal and ventricular ectopic beats more than reliable coverage of all five classes. Controlled tests of rhythm context, the second ECG channel, and minority-class learning are needed before the model is evaluated as a candidate for broader clinical data.

CRediT authorship contribution statement

Alireza Abbaszadeh: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration. Vahid Torkzadeh: Supervision, Writing – review & editing.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

Declarations of interest: none.

Ethics approval and consent to participate

Ethics approval and informed consent were not required because this study used the publicly available, fully de-identified MIT-BIH Arrhythmia Database. No new human or animal data were collected.

Data Availability

The MIT–BIH Arrhythmia Database is openly available from PhysioNet at PhysioNet MIT–BIH Arrhythmia Database v1.0.0. Source code, experiment configurations, trained models, and the prediction arrays used for the analyses are available in the public ECG Heartbeat Classification GitHub repository. The exact reproducibility snapshot used for the reported results is preserved in the immutable v1.0-joe-submission software release.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI ChatGPT (GPT-5.6 Sol) to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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